

STATISTICS

PART 2: Measures of Central Tendency, Measures of Variability

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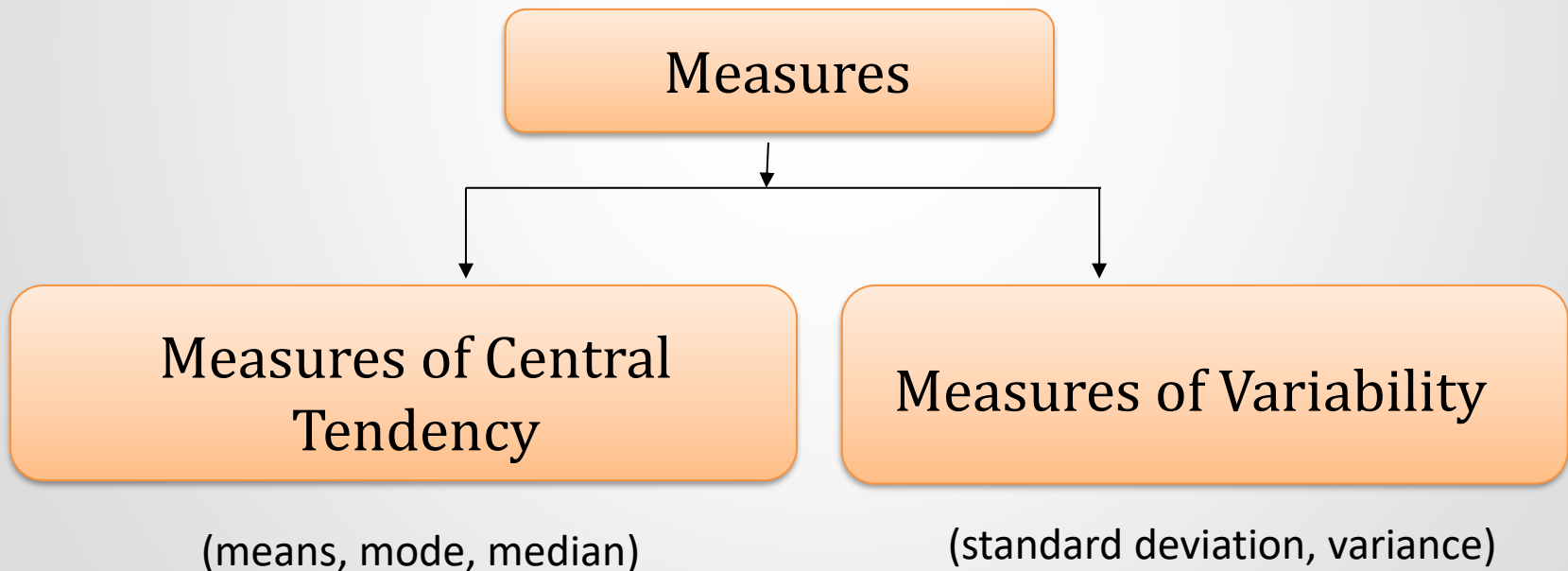
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Numerical Measures

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A data set can be described (summarized) by using graphics and tables as we saw before. A more efficient way of summarizing the data is using numerical measures.



Measures of Central Tendency

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Measures of Central Tendency

A) Means:

1) Arithmetic Mean: $\bar{x}_a = \frac{1}{n} \sum_{i=1}^n x_i$

$$\bar{x}_a = \frac{\sum_{i=1}^k f_i x_i}{\sum_{i=1}^k f_i}$$

Example:

Midpoints (x_i)	Frequency (f_i)
2	8
3	3
4	1

If data are given by frequency distribution (or it is classified)...

$$\bar{x}_a = \frac{2 \times 8 + 3 \times 3 + 4 \times 1}{8 + 3 + 1} = \frac{29}{12} \cong 2.417$$

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Measures of Central Tendency

A) Means:

2) Geometric Mean:

$$\bar{x}_g = \left(\prod_{i=1}^n x_i \right)^{\frac{1}{n}}$$

$$\bar{x}_g = \left(\prod_{i=1}^k x_i^{f_i} \right)^{\frac{1}{\sum_{i=1}^k f_i}}$$



If data are given by frequency distribution (or it is classified)...

Example:

Midpoints (x_i)	Frequency (f_i)
2	8
3	3
4	1

$$\bar{x}_g = \sqrt{(8+3+1) 2^8 3^3 4^1} \cong 2.345$$

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Measures of Central Tendency

A) Means:

3) Harmonic Mean:

$$\bar{x}_h = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}$$

$$\bar{x}_h = \frac{\sum_{i=1}^k f_i}{\sum_{i=1}^k \frac{f_i}{x_i}}$$



If data are given by frequency distribution (or it is classified)...

Example:

Midpoints (x_i)	Frequency (f_i)
2	8
3	3
4	1

$$\bar{x}_h = \frac{8 + 3 + 1}{\frac{8}{2} + \frac{3}{3} + \frac{1}{4}} \cong 2.286$$

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Measures of Central Tendency

A) Means:

4) Quadratic Mean:

$$\overline{x}_q = \left(\frac{1}{n} \sum_{i=1}^n x_i^2 \right)^{1/2}$$

$$\overline{x}_q = \left(\frac{\sum_{i=1}^k f_i x_i^2}{\sum_{i=1}^k f_i} \right)^{1/2}$$

If data are given by frequency distribution (or it is classified)...

Example:

Midpoints (x_i)	Frequency (f_i)
2	8
3	3
4	1

$$\overline{x}_h = \sqrt{\frac{8 \times 2^2 + 3 \times 3^2 + 1 \times 4^2}{8 + 3 + 1}} \cong 2.5$$

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Measures of Central Tendency

A) Means:

For any data set, the means satisfy the following relation:

$$\overline{x_h} \leq \overline{x_g} \leq \overline{x_a} \leq \overline{x_q}$$

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B) Median (Me):

To determine **median**, firstly put the numbers in order. Then,

If n is odd, the median is the middle number.

If n is even, the median is the average of two middle numbers.

Example:

$$X = \{2, 3, 1, 9, 2, 3, 4\}$$

$$1, 2, 2, \mathbf{3}, 3, 4, 9$$

$$Me = 3$$

Example:

$$X = \{7, 16, 40, 35\}$$

$$7, \mathbf{16}, \mathbf{35}, 40$$

$$Me = \frac{16+35}{2} = 25.5$$

Measures of Central Tendency

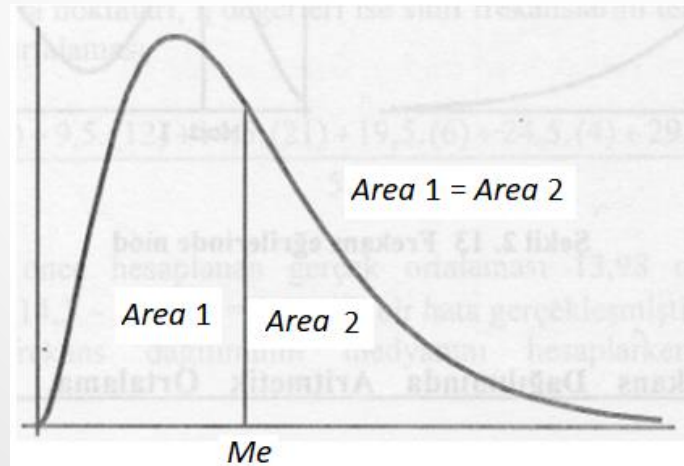
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B) Median (Me):

Consider the following frequency curve.



The median divides the data into two equal parts.

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B) Median (Me):

Median for the classified frequency distribution (CFD) is computed as:

$$Me = L + \left(\frac{n/2 - (\sum f)_2}{f_{median}} \right) c$$

L : The lower bound of the median class (the class containing the median)

n : The number of the data

$(\sum f)_2$: The sum of the frequencies of the classes that are smaller than the median class

f_{median} : The frequency of the median class

c : The class interval of the median class

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C) Mode(Mo):

The most frequently occurring number in a data set is called the **mode** of the data.

Examples:

$X = \{2, 3, 3, 3, 4, 4, 5, 5, 5, 5, 6, 7, 8\}$ $\Rightarrow Mo = 5$

$X = \{2, 3, 5, 7, 9, 10\}$ $\Rightarrow$ No Mode

$X = \{2, 3, 3, 3, 4, 4, 5, 5, 5, 6\}$ $\Rightarrow Mo = 3 \text{ and } 5$

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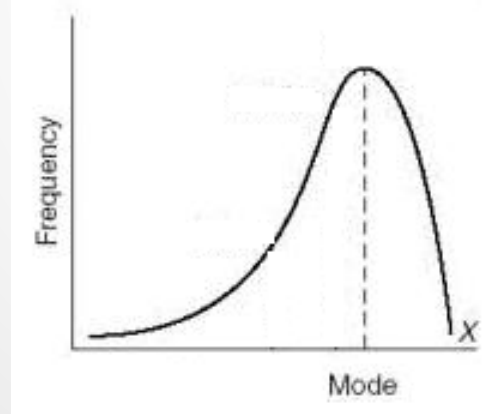
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C) Mode(M_o):

Consider the following frequency curves.



Mode is the point on the x-axis that corresponds to the maximum height of the curve.

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C) Mode(Mo):

Mode for the CFD is computed as:

$$Mo = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} c$$

L : The lower bound of the mode class (the class containing the mode)

c : The class interval of the mode class

$\Delta_1 =$

frequency of the
mode class

—

frequency for the previous
class of the mode class

$\Delta_2 =$

frequency of the
mode class

—

frequency for the next class
of the mode class

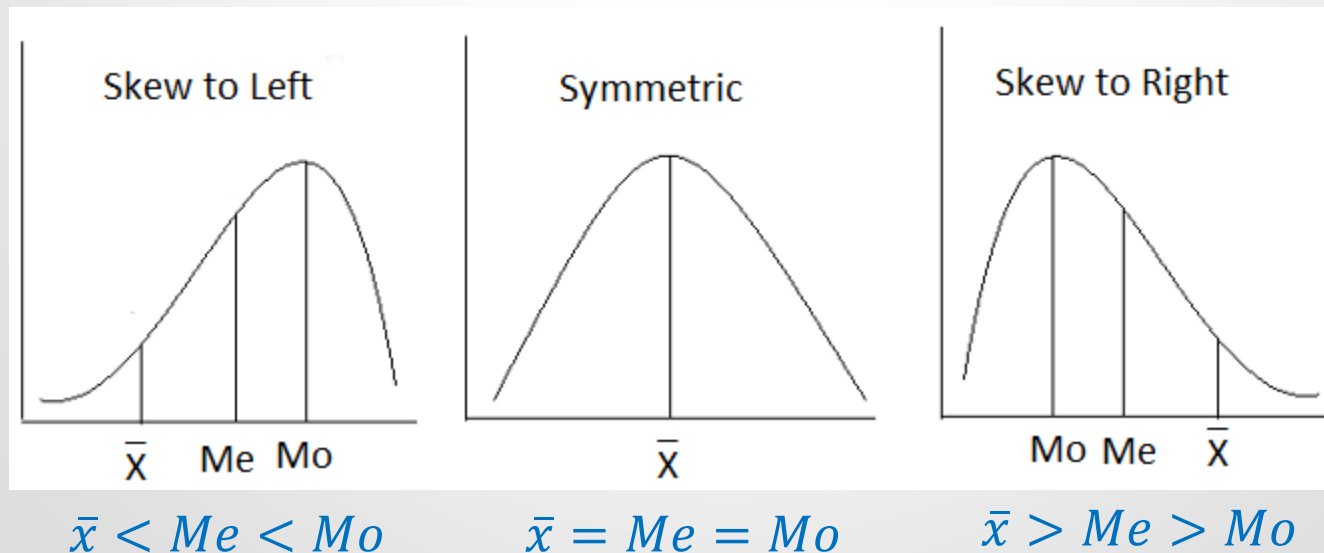
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Measures of Central Tendency

- Arithmetic Mean → The most important and used measure
- Median → Not sensitive to extreme values compared with arithmetic mean
- Mode → Can be also used for categorical data



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Measures of Central Tendency

Example: Consider the **PUZZLE EXAMPLE**.

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)
2-7	4,5	5
7-12	9,5	12
12-17	14,5	21
17-22	19,5	6
22-27	24,5	4
27-32	29,5	2

- Compute arithmetic mean for the exact and classified data. Find the error.
- Compute median for the exact and classified data. Find the error.
- Compute mode for the exact and classified data. Find the error.

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Measures of Central Tendency

Example:

Classes (C_i)	Frequency (f_i)
0-5	1
5-10	2
10-30	3
30-50	4

For the given frequency distribution table, determine the median

- by using formula.
- by using increasing (or decreasing) cumulative frequency polygon.

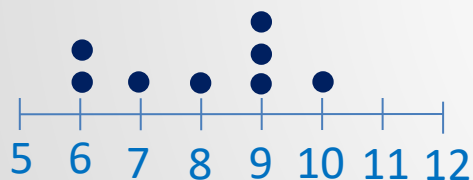
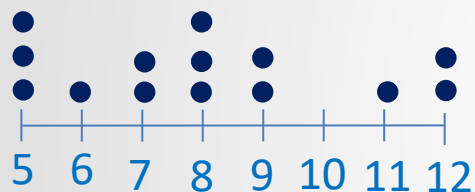
Measures of Variability

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Measures of Variability

Consider the following dot plots:



For each distribution, $\bar{x} = 8$. Their mean is the same; however, their distribution (variation) is different as seen clearly. For this reason, we need an extra measure to show this difference.

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Measures of Variability

Consider the data set:

$$X = \{x_1, x_2, x_3, \dots, x_n\}$$

↓ Deviations from the arithmetic mean

$$X = \{x_1 - \bar{x}_a, x_2 - \bar{x}_a, x_3 - \bar{x}_a, \dots, x_n - \bar{x}_a\}$$

If we use the arithmetic mean to summarize this new data...

$$\sigma_a = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}_a) = \frac{1}{n} \sum_{i=1}^n x_i - \frac{1}{n} \sum_{i=1}^n \bar{x}_a = 0$$

This measure can not be used since it is always 0.

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Measures of Variability

The quadratic mean is a more appropriate mean to summarize the given deviations.

$$\sigma_x = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

This measure is called **standard deviation**. For the frequency distribution,

$$\sigma_x = \sqrt{\frac{1}{\sum_{i=1}^k f_i} \sum_{i=1}^k f_i (x_i - \bar{x})^2}$$

Unless otherwise stated, we assume that mean refers arithmetic mean, i.e., $\bar{x} = \overline{x_a}$.

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Measures of Variability

The square of the standard deviation, namely σ_x^2 , is called **variance**.

Theorem:

For the exact data,

$$\sigma_x^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2$$

For the frequency distribution,

$$\sigma_x^2 = \frac{1}{\sum_{i=1}^k f_i} \sum_{i=1}^k f_i x_i^2 - \bar{x}^2$$

Prove them!

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Measures of Variability

Example: Consider the **PUZZLE EXAMPLE**. Compute variance for the exact and classified data. Find the error.

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)
2-7	4,5	5
7-12	9,5	12
12-17	14,5	21
17-22	19,5	6
22-27	24,5	4
27-32	29,5	2